Project Demonstration

Scalability & Future Plan

Date	03 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

This document outlines how the current OptiCrop solution can be scaled to handle larger user bases, increased data loads, or extended features in the future, and captures the team's roadmap for enhancing the project beyond its current state.

Current System Limitations

S.No	**Limitation**	**Impact**	**Priority to Address (High/Medium/Low)**
1	Single-instance Streamlit app with no database	Cannot store user history or usage analytics	Medium
2	Model trained on a fixed 2200-record dataset	May not generalize perfectly to regions/soil types not well represented in the training data	Medium
3	No authentication layer	Cannot personalize recommendations per user or restrict access	Low
4	English-only interface	Limits accessibility for non-English-speaking farmers, the primary target user	High

Scalability Plan

S.No	**Scalability Aspect**	**Current State**	**Proposed Upgrade / Solution**

1	User Load	Single free-tier Streamlit Cloud instance	Migrate to a paid tier or containerized deployment (e.g., Docker + cloud hosting) if concurrent usage grows
2	Data Storage	No persistent storage; stateless per-session predictions	Add a lightweight database (e.g., SQLite/PostgreSQL) to log predictions and enable usage analytics
3	Performance	Model loads fresh on cold start (~2-3 sec delay)	Cache the loaded model in memory across sessions using Streamlit's @st.cache_resource
4	Security	No authentication; public tool	Add optional login for farmers who want saved recommendation history

Future Roadmap

Phase	**Planned Feature / Enhancement**	**Target Timeline**	**Expected Impact**
Phase 2	Multi-language support (Telugu, Hindi, and other regional languages)	Next 3 months	Makes the tool accessible to a much larger base of rural farmers
Phase 3	Live weather API integration for automatic climate input	Next 6 months	Removes manual entry of temperature/humidity/rainfall, improving ease of use
Phase 4	Fertilizer and irrigation recommendations alongside crop suggestions	Next 9-12 months	Extends the tool from crop selection into full crop-management guidance